HACKERING 2.0 HACKATHON | SOFTWARE TRACK: REAL-TIME VOICE AI

NeuroVocal AI: Dual-Stream Voice Triage & Acoustic Biomarker Engine

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What is this project about? Most emergency voice chatbots only transcribe spoken words into text. However, during life-threatening medical emergencies like strokes, respiratory failure, or shock, callers are often confused, panicking, or physically slurring. An elderly person might simply say, *'I just feel a little tired,'* while acute facial paralysis is already setting in. A standard chatbot suggests rest. **NeuroVocal AI** listens to how the caller's body physically sounds while speaking—turning standard microphone audio into a real-time digital stethoscope that detects acute stroke, vocal fold paresis, and hypoxia at 25 Hz without needing extra hardware.

1. System Architecture: Two Streams Running in Parallel

Instead of forcing a large language model to guess medical status from transcripts, we separated the system into two asynchronous tasks running simultaneously off the same 16 kHz audio buffer:

Stream 1: Conversational Operator (Front-of-House)	Stream 2: Digital Signal Processing Scanner (Background)
1. Deepgram Nova-2 handles fast speech transcription. 2. Groq Llama 3.3 (70B) maintains a calm, structured triage protocol, asking for caller location and keeping them still. 3. Cartesia Sonic synthesizes clear, empathetic voice audio. *Purpose:* Reassure the patient and collect critical dispatch details.	1. Captures raw PCM audio in a dedicated circular ring buffer. 2. Analyzes sliding 64 ms audio frames every 40 ms (25 Hz). 3. Computes pitch, jitter, noise ratio, and vocal tract formants using pure mathematical DSP (no LLM latency). *Purpose:* Detect physical symptoms before the caller can explain them.

2. The Acoustic Biomarkers We Implemented

We implemented four clinical speech biomarkers in Python (neurovocal_poc.py) using NumPy and SciPy. Each biomarker targets a specific physical symptom:

Biomarker	How We Compute It	Healthy Range	What Abnormal Values Mean
Fundamental Pitch (F0)	Normalized Autocorrelation with parabolic interpolation	85 to 255 Hz	Rapid flutter indicates autonomic panic, shock, or breathing distress.
Jitter (Perturbation)	Period perturbation quotient across pitch cycles	Under 1.04%	Jumps above 2.50% when stroke or facial weakness causes vocal fold asymmetry.
Harmonics-to-Noise (HNR)	Ratio of periodic voice energy to residual noise (dB)	Over 18.0 dB	Drops below 12.0 dB when turbulent breathiness replaces clean phonation (hypoxia).
Formants (F1, F2)	14th-order Linear Predictive Coding (Levinson-Durbin)	F1: 300-900 Hz F2: 900-2400 Hz	In acute stroke (dysarthria), vowel space collapses and F2 centralizes to 1500 Hz.
CSI Composite Score	Weighted index: 0.40(Jitter) + 0.35(Formants) + 0.25(HNR)	0.00 to 0.25	Score >= 0.65 immediately triggers an emergency 'Code Red: Suspected Stroke' alert.

3. Real Engineering Benchmark & Latency Results

For a real-time emergency dashboard, latency buildup is unacceptable. If a network hiccups, the server must never queue stale audio packets. We tested our FastAPI server (server.py) by collecting 100 consecutive frames at 25 Hz:

Benchmark Test Parameter	Target Design Value	Measured Real Result	Evaluation Verdict

Telemetry Broadcast Rate	25.0 Hz (every 40 ms)	25.17 Hz	**PASSED**
Inter-Packet Timing	40.0 ms per packet	40.00 ms (std dev: 0.54 ms)	**PASSED**
Packet Schema Integrity	100% (all 9 metrics present)	100 / 100 packets verified	**PASSED**
Clock Drift / Latency Buildup	Under 50.0 ms total drift	0.62 ms over 4.0 seconds	**PASSED (Zero Latency)**

4. How Judges Can Test Our System Live

Quick Setup (Runs on any laptop):

1. Install dependencies: pip install -r requirements.txt
2. Launch backend: python server.py (runs on port 8000)
3. Open the dispatcher HUD: Go to http://localhost:8000/client in your browser.
4. **Try the Stroke Demo:** Click the *'Stroke / Dysarthria'* button. Watch the CSI score rise to 0.80 and the flashing **CODE RED: ACUTE STROKE BIOMARKER DETECTED** banner appear.
5. **Try Live Mic:** Click *'Live Hardware Mic'* and speak to see your real pitch and vocal resonances graph at 25 Hz.
6. Run the automated timing test: python test_client.py